Beyond Linear Cointegration: Regime-Switching and Tail Dependence in BWP/USD and ZAR/USD Exchange Rates

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Introduction

- Introduced in 1976, the Botswana Pula (BWP) was pegged into a basket of currencies such as the United States Dollar (USD) as one of Botswana's trading partners Thupeng (2021).
- As commodity prices surged, diamonds being Botswana's main commodity allowed for the Pula to strengthen against the USD.
- However in periods such as the 2007-2009 global recession lead to investors seeking refuge to the USD due to the global decline in demand of commodities.
- Along with the COVID-19 pandemic lead to high foreign exchange rate volatility among global financial markets.
- Maveé et al. (2016) highlights factors influencing exchange volatility which include, economic surprises domestic and in the United States, fluctuations in commodity prices and political uncertainties to name a few.

Introduction cont...

- For both the Pula and South African Rand (ZAR) influences such as economic situations, market sentiments geopolitical events and external shocks have an effect on their volatility and price movements.
- Key indicators of the Pula are inflation rates, unemployment figures and gross domestic product (GDP) depict the strength of the currency.
- Relationships between the BWP and ZAR differ due to their nature, market behaviour and external differences Israel (2023).
- With the BWP heavily reliant on diamond exports Botlhale (2021) and domestic inflation rates and monetary policies issued out by the Bank of Botswana.

Introduction cont.

- As such, most empirical studies base their findings on macroeconomic variables or stock datasets may have a linear relationship, which may not be the case due to an exhibiton of a non linear relationship Makatjane et al. (2017).
- The handling of such datasets must be carefully considered in order not to lead to misleading findings and policy recommendations.
- Modelling such non linear dynamics introduces a Markov switching cointegration model, specifically a Markov switching vector error correction model (MS-VECM) will help in sound decision making.
- MS-VECM will help to establish cointegration and account for short term deviations and switching behaviour Mahmoudi and Ghaneei (2022).

Problem Statement

- Granger's cointegration proposed by Granger and Lee (1989), are widely used to determine two or more time series are cointegrated despite non stationarity in their levels.
- This particularly involves pair-wise time series and assumes a linear relationship between the time series.
- This is seen as a limitation to such a procedure particularly if the time series exhibit an non linear relationship.
- As Granger's cointegration may lead to misrepresented results and ineffective capturing of actual dynamics between time series Castle and Hendry (2017)
- Thus the introduction of the Johansen cointegration procedure which handles multiple time series simultaneously Sekakela (2019).

Problem Statement cont...

- Johansen's procedure views the cointegration relationship as part of a larger system of equations allowing for estimation of multiple cointegrating vectors among variables.
- However, the procedure does have limitations such as Caporale et al. (2021) instigated such as having the same integrated order I(1), this involves non stationary variables at their levels to be differenced once to achieve stationarity.
- Failure to achieve integration of the same order leads to inaccuracies and the invalidation of the Johansen cointegration method.

Research Objectives

Primary Objective

To investigate the regime switching cointegration between the BWP/USD and ZAR/USD exchange rates and to further assess their dependence.

As such secondary research objectives as solutions have been formulated as:

- To estimate the MS-VECM in order to capture the long run regime switching relationship between BWP/USD and ZAR/USD.
- To estimate the Clayton copula in order to caputure dependency between BWP/USD and ZAR/USD.

Significance of Study

- The findings of this study will help understand and capture long term relationships between the BWP/USD and ZAR/USD exchange rates, thus, cointegration.
- Lastly, with the help of the Clayton copula the capturing of dependency structure amongst the exchange rates will help with policy recommendations, see Molefe and Makatjane (2020).

Literature Review — Key Insights

- **Linear vs Nonlinear Cointegration**
  - Traditional cointegration assumes linear adjustment (Granger & Lee, 1989).
  - Evidence shows exchange rates have **asymmetry, structural breaks, and nonlinearity** (Haug & Basher, 2011; Dufrénot & Mignon, 2012).
  - Nonlinear cointegration performs better during high volatility.
- **Regime-Switching Models**
  - Markov-Switching models (Hamilton & Raj, 2002/2013) identify **stable vs crisis** regimes.
  - MS-VECM captures regime-dependent short-run adjustments (Phoong & Sek, 2013).
- **Copula-Based Dependence**
  - Correlations fail to capture asymmetric dependence.
  - Clayton copula models **lower-tail dependence**—joint sharp depreciations (Patton, 2001; Liu et al., 2020).

Methodology — 1 of 3

Our preliminary analysis includes the following assessments into the distribution of the two exchange rates as follows:

- **Normality Tests** includes:
  - Skewness and Kurtosis tests - asymmetric tails from Normal distribution
  - Jarque-Bera test
  - Anderson-Darling test
- **Tests for highly correlated returns**
  - Kendall's tau - determine monotonic dependence
  - Spearman's rank correlation coefficient- monotonic relationships
  - Johansen cointegration test: Max eigenvalue test - presence of one cointegrating factor

Nonlinear and Nonlinear unit root tests

- White's Neural Network test : Non linearity by an undefined conditional expectation function.
- Kapetanios Shin and Shell test : capture a nonlinear adjustment function
- Perron's nonlinear unit root test: Detect unit roots in structural breaks

Methodology - 2 of 3

- **MS-VECM** captures:
  - Long-run equilibrium by a cointegrating vector β
  - Regime-dependent short-run adjustments,$\alpha(s_t)$
  - Short term regime dependent parameter $\Gamma(s_t)$
- **Regimes governed by a Markov chain:**

$$P(s_t = j | s_{t-1} = i) = \begin{bmatrix} p_{11} & p_{12} & \cdots & p_{1j} \\ p_{21} & p_{22} & \cdots & p_{2j} \\ \vdots & \vdots & \cdots & \vdots \\ p_{i1} & p_{i2} & \cdots & p_{ij}. \end{bmatrix}$$

- **Model Selection:**
  - Likelihood Ratio test: rejection of a single regime at 5% significance level.
  - Two-regime structure adopted (stable vs. crisis/high volatility)
- **Parameter Estimation:**
  - EM algorithm using the Hamilton filter for regime probabilities
  - Parameters updated per regime: $\alpha(s_t), \Gamma(s_t), \beta$

Methodology - 3 of 3

Dependence Measures:

- Kendall's tau
- Spearman's rank correlation coefficient
- Clayton Copula
- Lower-tail dependence - measure of dependence during episodes of shocks and crises

Exchange rate movement - 01/01/2015 to 14/11/2024

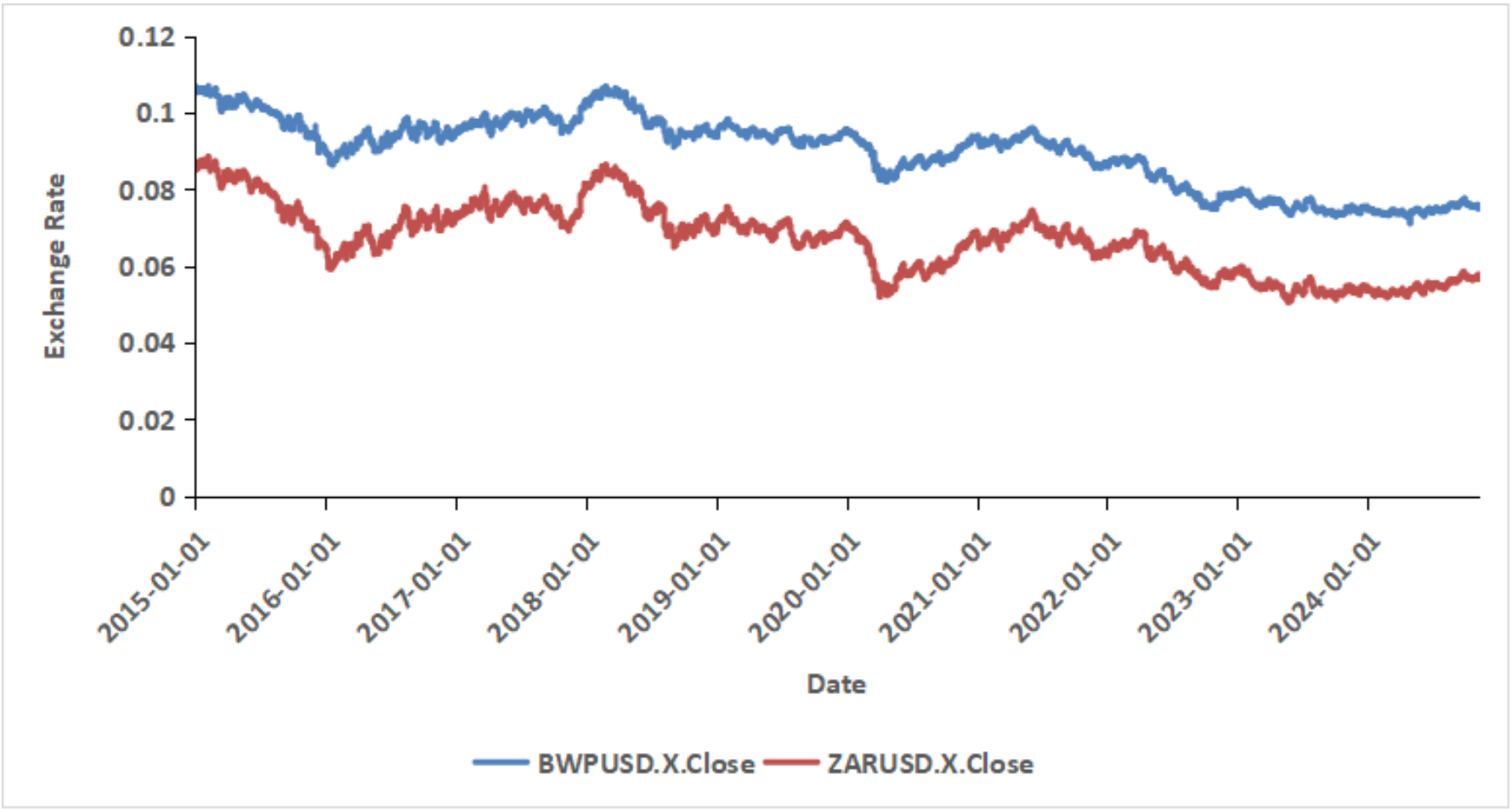


Figure: Nine year closing exchange rate data of BWP/USD and ZAR/USD

- Overall decline in exchange rates since 2018.
- The movement of the two exchange rates against the USD is similar from 2015 until 2020. 2020 BWP Decline: In 2020, the BWP experienced a less significant decline compared to the ZAR during the COVID-19 pandemic.
- Reasons for BWP Stability: This relative stability is attributed to Botswana's less diversified market and its reliance on diamond trading (Thupeng, 2021).
- ZAR Volatility: The ZAR, being a floating currency with high volatility against the US Dollar, was more susceptible to global shocks, as emphasized by Masuku (2016).
- Post-Lockdown Divergence (2021): Following the easing of lockdown measures in 2021, the ZAR's rate increased, while the BWP's strengthening against the USD was more gradual.
- Factors Impacting ZAR: The ZAR's performance as a floating currency was significantly influenced by the pandemic, global economic shocks, and even specific events like claims of weapon trades with Russia Ryan (2023).

Results and Discussion – Page 1 of 2

Table: Normality, Highly correlated returns and Cointegration Test Results

Category	Test	BWP/USD	ZAR/USD	p-value
Normality	Skewness	-0.4086	0.1252	–
	Kurtosis	-0.9169	-0.7760	–
	Jarque-Bera	161.93	71.37	< 0.001
	Anderson-Darling	4818.1	4674.6	< 0.001
€orrelation and Cointegration	Kendall's τ	0.8858		–
	Spearman's Rank	0.9810		–
–	Johansen (Max Eigenvalue)	0.0079		< 0.001

- Similar platykurtic distribution like Aliyu et al. (2023) use of the Nira against USD.
- Asymmetry to the normal distribution using Anderson-Darling and Jarque-Bera similar to Hamilton and Raj (2013) and Makatjane (2024).
- Presence of at least one cointegration factor.

Results and Discussion – Page 2 of 2

Table: White Test, KSS Test, and Phillips-Perron Results

Category	Test	BWP/USD	ZAR/USD	Value/Result	p-value
White NN Test	F-Statistic	5.844	10.232	df = (2, 2573)	0.0029 / <0.001
KSS Nonlinear	Lag	-0.0266 (SE=0.0147)		t = -1.81	0.0703
	Lag.1	-0.0571 (SE=0.0139)		t = -4.10	<0.001
Phillips-Perron	Test Statistic	-2846.5	-2474.5	Lag = 9	0.01
	Alternative Hyp.	Stationary	Stationary	–	–

- White's neural network test significant for both exchange rates as non linear Makatjane et al. (2017).
- The KSS non-linear lag represents the BWP/USD and ZAR/USD lagged regression model. Lag.1 represents the lagged regression model differenced once.
- The KSS tests using the student's t-test has proved to be significant in investigating stationary among non linear series, Liu and He (2010) and Chang et al. (2015).
- Phillips-Perron test proved to be significant in determining stationarity by first order differencing.

Markov switching error correction model (MS-VECM)

Table: Likelihood ratio (LR) Test Results for *BWPUSD* and *ZARUSD*.

Series	**Test Statistic**	**Degrees of freedom (df)**	**p-value**
BWPUSD and ZARUSD	348.7424	7	< 0.001

- The test was formulated across the null hypothesis being that there is only one regime against the alternative being that there is at least two regimes at 5% significance level Qu and Zhuo (2021).
- The null hypothesis is rejected and there is sufficient evidence to justify the use of two regimes for the multivariate time series of the exchange rates Makatjane (2024).

Transition probability matrix

- The transition probabilities were given as:

$$p_{ij} = \begin{bmatrix} 0.5933 & 0.4067 \\ 0.5895 & 0.4105 \end{bmatrix}$$

- In regime 1, p_{11}, there is a 59% chance exchange rates stay stable and in p_{22} a 41% chance they'll shift into periods of high volatility and crises in regime 2 .
- This level of regime switching indicates that the exchange rates are dynamic in changing from stable conditions to experiencing episodes of crises and shocks on exchange rates Chkili and Nguyen (2014).

Parameter Estimation

Table: Loading coefficients indicate adjustment speed towards equilibrium.

Equation	Coefficient	Std. Error	z-value	$P > \|z\|$
BWPUSD=X (ec1)	0.0008	0.002	0.362	0.717
ZARUSD=X (ec1)	0.0101	0.003	3.268	0.001

$$\beta = \begin{bmatrix} 1.0000 \\ -1.3500 \end{bmatrix}$$

The cointegration relation vector β for the two currencies suggests that the formulation of a cointegrating equation can be represented as:

$$\epsilon_t = BWPUSD - 1.35 ZARUSD \tag{1}$$

Dependence measures

Table: Results of dependence measures between the exchange rates

Kendall's τ	Spearman's ρ	Clayton copula (θ)	Lower tail dependence (λ_L)
0.355	0.524	1.1016	0.710

- Positive association among the exchange rates. Clayton copula indicated a strong dependence.
- The high value of the lower tail dependence suggests that in times or periods of economic volatility and shocks the two exchange rates have a very high tendency to move together Chlond (2018).

Remarks

- Exchange rates proved to be asymmetrical, non-linear, highly correlated and stationary when differenced once.
- A two regime was obtained by the likelihood ratio test.
- Transition probability matrix which indicates a dynamic regime switching between the two exchange rates.
- Long-term cointegration relation of $BWP = 1.35ZAR$.
- The dependency of both exchange rates is relatively high as they influence one another during periods of uncertainty.

Conclusion

- Normality tests, White Neural Network, KSS and Perron tests confirmed **nonlinearity**.
- Evidence of **cointegration** between BWP/USD and ZAR/USD (Johansen test).
- MS-VECM identified **two regimes** with frequent switching:
  - Regime 1: Stable market conditions
  - Regime 2: Crisis / high volatility
- Adjustment toward long-run equilibrium is **faster in crisis periods**.
- Clayton copula revealed **strong lower-tail dependence**: currencies depreciate jointly during market stress.

Conclusion — Recommendations & Future research

- Implication for policymakers:
  - Need for **coordinated regional monitoring** of exchange rate risks.
  - Improved stress testing using copula-based tail-risk models.
- Contribution:
  - Combines **regime-switching cointegration and tail dependence**, enriching the literature applied to Southern African currencies.
  - Offers a framework for understanding exchange rate dynamics in Southern Africa.

Future research

- Neural network-based models to further explore nonlinear regime-switching cointegration.
- Further insights into copula families
- Comparisons involving tail-heavy distributions such as the Student's t-distribution.

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